Efficient Fine-Tuning of Vision Transformer for Pneumonia Detection: A Partial Freezing Approach

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Abstract*—Pneumonia, a world's leading infectious disease causing deaths, causes around 2.5 Million deaths annually. In this research paper a complete pipeline of supervised Deep learning for automated pneumonia detection from the Chest X-Ray images is presented using a Vision Transformer ViT. We carried out the study in three stages: 1) Developed a research proposal to identify the Vision Transformer as a perfect architecture for this particular task, 2) Using Singh et al. (2024) model as a baseline to reproduce the results, achieving an AUC-ROC of 0.97 accuracy of 0.9167 on a dataset: Kermany chest X-Ray, 3) Introduced partial fine tuning using the concepts of layer freezing and applying label smoothing for regularization, lastly fixing the data leakage problem in the validation part to extend our approach. In the last stage we froze 10/12 ViT encoder blocks to achieve a reduction of 84% trainable parameters by reducing the number from 86 million to 14 million. In this way the collapse during full fine tuning was avoided and training was stabilized. Extending our approach, we tested the model on BloodMNIST dataset for multi-class blood cell classification where we were able to achieve an AUC-ROC of 0.9917 and accuracy of 0.9030. from this we deduced that partial fine tuning along with appropriate regularization can provide a more stable and efficient strategy for adapting large pre-trained vision models in those areas where there is a restricted or bounded medical data available

***Index Terms*—Pneumonia Detection, Vision Transformer, Kermany Chest X-Ray, Fine-Tuning, Freezing Layers**

I. Introduction

Pneumonia is a dangerous infectious disease affecting people all around the world. According to the World Health Organization it is the one of the major reasons of the death of children younger than five. Pneumonia is when the lungs get inflamed. This is usually caused by bacteria, viruses and fungi. Doctors use something called a chest X-ray to figure out if someone has pneumonia. This is a way to do it because X-ray machines are easy to find and do not cost too much. It is not easy to look at a chest X-ray and know for sure if someone has pneumonia. It takes a lot of practice to get good at it. Experienced doctors who look at X-rays all the time can miss signs of pneumonia that are not easy to see. In some countries where pneumonia is a big problem there are not enough doctors to look at all the X-rays. One way to make things better is to use computers to help diagnose pneumonia. In order to find out what to look in an X-Ray Image, these computers try to learn by teaching themselves. They can do this on their own without being told what to look for. In the middle of the 2010s something called Convolutional Neural Networks started being used to look at pictures. These computers got really good at looking at pictures and figuring out what was wrong with people like if they had something, with their eyes or skin or if they had pneumonia from a chest X-ray. Pneumonia diagnosis is getting better with the help of these computers. Pneumonia is a problem and these computers are helping to solve it. But the main problem is that Convolutional Neural Networks (CNNs) have some limitations as they only capture a small region of the image at a time not the whole. And we know that in medical imaging it is very important for a model to capture long range dependencies to avoid misclassification. For example, the lobar pneumonia showing a two sided appearance or slight interstitial changes across the lung may be overlooked. Vision Transformers (ViTs), as described by Dosovitskiy et al. (2021) in "An Image is Worth 16x16 Words", overcome this problem by using the Transformer's self-attention mechanism in combination with global images. The model treats an image as a sequence of fixed-size image patches, embeds them linearly in a vector and processes the sequence using standard Transformer encoder layers with multi-head self-attention to allow direct modelling of global dependencies.

Because each attention head computes the interactions of each token pair, the self-attention can relate a patch to every other patch regardless of its position. The large receptive field is ideal for chest X-ray interpretation: pneumonia is characterized by diffuse opacities that can affect multiple anatomical regions and its detection depends on the simultaneous consideration of information from the entire radiograph.

This paper reports the full documentation of a project on pneumonia detection, in three phases. In Phase 1; A Research Proposal was submitted that identified ViT-Baseline-Patch16-224 as our target Architecture and the Kermany dataset as our target for this project. In Phase 2, we reproduced the approach of this award-winning paper which uses Vision Transformers (ViT) to detect pneumonia and achieved 91.67% test accuracy and 0.97 AUC-ROC, even after facing a disastrous collapse: where all 86 million ViT parameters are updated for every training image (a total of around 4,172 images), the model collapses in training with training loss approaching zero (0.00) and training F1 approaching 1.0 (perfect), potentially due to memorization of the training dataset. In Phase 3, three different principled interventions were proposed and tested to prevent such failure mode, and is the core contribution of this paper.

The three interventions are:

- **Layer Freezing:** The first layer (Patch Embedding), and the first ten out of twelve (ViT Encoder Blocks) were frozen so we could avoid Over-fitting to the Limited Dataset.
- **Label Smoothing:** One hot Target Labels of Cross Entropy Loss will be replaced with Soft Labels (ϵ = 0.1) so we can avoid overly confident predictions, and Regularize Model Output.
- **Validation Transform Patch:** This patch repaired a Data Leakage Problem within the Validation Dataset, which used the Transform Layers from the Training Dataset thus losing an Evaluation/Model Selection Signal.

The effects of these changes are shown through the use of the Original Kermany Pneumonia Dataset and another Blood Cell Classification dataset (BloodMNIST) on eight classes from the MedMNIST Benchmark. The rest of this paper is organized as follows. Section 2 provides a review of the Literature Related to this Project. Section 3 includes details of the new Model along with Proposed Changes.Experimental Settings for our project are outlined in Section 4. Results for our project are presented in Section 5. We Analyze and Discuss the Results in Section 6. Finally, we Conclude in Section 7

The main purpose for this proposal was to show that models that are purely transformer based with no convolutional bias, are the best for image classification but the data on which they are trained should be sufficient enough.

II. Background and Related Work

A. Architecture of Vision Transformer

The concept of the Vision Transformer (ViT) was proposed by Dosovitskiy et al. (2021). The main purpose for this proposal was to show that models that are purely transformer based with no convolutional bias, are the best for image classification but the data on which they are trained should be sufficient enough. Specifically, this work adopts the model ViT-Base-Patch16-224, which breaks down the image into 196 patches of 16 x 16 pixels from the original 224 x 224 image. The patch is converted from 256-dimensional pixels to 768-dimensional vector via a learnable linear transformation.

Each vector is also combined with a learnable positional encoding that represents the position of the patch. We prepend a special learnable token, [CLS], or classification token, to the sequence of patches to get a total of 197 tokens. These tokens are then fed to a stack of 12 encoder layers of transformer. These layers have two sub-layers, which are connected by a residual connection and followed by a layer normalization.Multi-head self-attention (MHSA), one of the two sub layers, with 12 attention heads each of dimension 64 (a total of 768 dimensions). The second sub-layer is a position-wise feed-forward network (FFN) with a hidden layer size of 3072, which also uses GELU.

And finally, after 12 layers, the last token [CLS] is used and processed by one more layer norm followed by a linear classification layer. The total number of parameters in ViT-B/16 is around 86 million.

B. Partial Fine-Tuning and Transfer Learning

The typical method of transfer learning to new downstream tasks using a small set of labeled data is to fine-tune a pre-trained, large-scale deep neural network. By default, all the weights of the pre-trained backbone are initialized with the pre-trained weights, and fine-tuned for the downstream task. It's a good strategy, but it's prone to overfitting if the new task's training set is small relative to the capacity of the network, and it can lead to deterioration of the pre-trained general visual features learned by the network during pre-training because of overfitting to the gradient descent updates.

To address these issues, selective fine-tuning (or layer freezing) freezes some layers at their initialized weights, and fine-tunes the rest. The rationale was given by Yosinski et al. (2014) who demonstrated that lower layers in deep visual feature networks learn general visual features that can be transferred to a new task (low-level texture and edge detectors useful for all kinds of visual tasks) whereas higher layers learn more task-specific features. This means that only the higher layers of a deep network should be fine-tuned to a new, downstream vision task and that freezing the lower layers can prevent overfitting and retain useful pre-trained features.

This strategy could also be applied to Transformer-based models: early encoder blocks will learn general patterns of token interactions and low-level features, while later blocks will learn higher-level features. By keeping most of the encoder frozen (e.g., the first 10 of 12 blocks), we effectively reduce the number of gradients that are updated during training, and hence reduce the capacity of the model to a level that matches the size of the downstream training set.

C. Label Smoothing as Output Regularization

Label smoothing is a side effect of the ImageNet Large Scale Visual Recognition Challenge by Szegedy et al. (2016) as part of the Inception architecture improvements to the ImageNet Large Scale Visual Recognition Challenge (ILSVRC).The traditional cross-entropy loss of a K-class classification problem is to have the network minimize the KL divergence between the predicted distribution and a one-hot empirical label distribution, making the probability of the ground-truth class in the predicted or output distribution to increase to 1.0. This encourages the network to produce logits of arbitrarily high value, which increases overconfidence.

Label smoothing counteracts this, and replaces the one-hot target distribution with a smoothed target distribution that is a combination of the one-hot target distribution and the uniform distribution. For example, in a two-class case, with ϵ=0.1, the target probability of the correct class is (1 - ϵ) + ϵ/K = 0.95, and the target probability of the incorrect class is ϵ/K = 0.05. This smoothed target enables the loss to be reduced to exactly zero and encourages large logit magnitudes to be small, and provides some mild regularization, which typically improves the model's generalization performance when the training data are small, and reduces the calibration error on held-out data.

The benefits of label smoothing have been verified by a number of subsequent studies. Muller et al. (2019) demonstrated that label smoothing improves the model calibration, a preferred feature of clinical decision support systems that explain the output probability of the model as confidence. Furthermore, Zoph et al. (2020) have adopted label smoothing in their EfficientNet training recipe.

D. Previous Pneumonia Detection Work

Chest radiography has been the subject of a large body of deep learning research. Kermany et al. (2018) proposed the benchmark dataset used for this work, as well as a transfer learning strategy with pre-trained CNNs, which achieved clinically meaningful results for pneumonia and retinal disease classification. Various CNN-based research has continued to improve this baseline through model architecture, data augmentation and ensembling.

The recent adoption of Transformer-based approaches in medical imaging was triggered by the success of global self-attention in encoding spread-out patterns in radiography. Singh et al. (2024), the baseline paper for this study, fine-tuned the ViT-Base-Patch16-224 model on the Kermany dataset and set a strong ViT baseline for this task by achieving an accuracy of 97.61%, sensitivity of 95%, specificity of 98%, and AUC-ROC of 0.96.

Hadhoud et al. (2024) developed a two-step CNN-ViT model for chest disease classification, which uses CNN feature extraction and Transformer self-attention, and reported a high multi-class accuracy on chest X-ray benchmarks. Alshanketi et al. (2024) focused on the role of class imbalance in pneumonia detection, developing a transfer learning approach with weighted sampling to manage class imbalance - a problem encountered in this work as well. Wang et al. (2021) benchmarked CNN and ViT models for medical image classification across several datasets, revealing that ViT models leverage substantial gains from pre-training on large-scale data, and also that partial fine-tuning approaches yield significant gains in performance when fine-tuning on downstream data with limited labels.

E. MedMNIST and BloodMNIST

MedMNIST v2, a set of large scale standardized benchmark for 2D and 3D biomedical image classification, was proposed by Yang et al. (2023) recently. This set includes 12 2D and 6 3D pre-processed datasets with multiple imaging approaches including, radiology, dermatology, microscopy, pathology, and ophthalmology. One of the 2D datasets, BloodMNIST, includes 17,092 microscopy images of peripheral blood cells in 8 morphologically different classes. This dataset is a subset of a larger blood cell dataset, and comes with pre-determined train/validation/test splits. MedMNIST offers a robust testing framework to evaluate the generalization ability of medical image models.

III. Proposed Model and Methodology

A. Baseline Architecture

Phase 2 is based on ViT-Base-Patch16-224 (loaded via HuggingFace Transformers) with ImageNet-21k pre-trained weights. A two-class linear head instead of the original 1,000-class head is used on the binary pneumonia detection task (two classes head on Kermany). Both the hidden and attention layers have dropout probabilities of 0.1. Initialization of the model is done using ignore mismatched sizes=True that re-initializes only the classification head but transfers 768-dimensional encoder weights in an appropriate manner. In Phase 2, all about 86 million of the parameters are trainable.

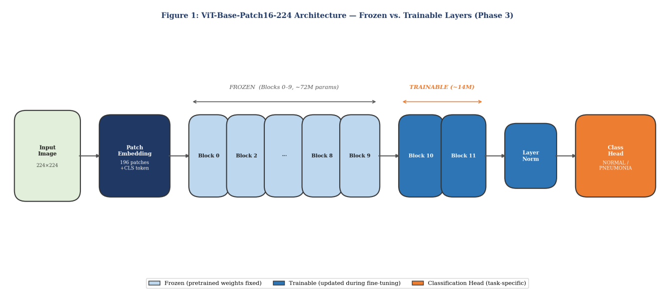

Fig. 1. ViT Architecture (Frozen vs Trainable).

B. Improvement 1: Layer Freezing for partial fine tuning

Fine-Tuning Partially through Frozen Layer. Phase 3 adopted the main architectural change as the use of partial fine-tuning by freezing layers. In particular, these groups of parameters are frozen (requires grad set to False) to their ImageNet pre-trained values:

- The patch embedding layer (vit.embeddings)
- Blocks 0-9 of 12 (vit.encoder.layer[0] through [9]) are blocked by ViT.

The groups of parameters below are trainable and updated at downstream fine-tuning:

- The last LayerNorm (vit.layernorm)
- The linear classification head (classifier)

In this way we can reduce the total number of trainable parameters from 86 million to around 14 million, which means a total of 84% decrease in the parameters. This is based on the hierarchical feature specificity of deep visual models defined by Yosinski et al. (2014): initial Transformer blocks encode low-level patch interactions, encoding the general visual primitives, such as edges and textures, that are universally applicable; middle blocks encode the shape and part-level features; and final blocks encode high-level semantic representations, which are task-specific. Freezing the initial 10 blocks does not change any of the general and mid-level representations used during pre-training, only the last two blocks and classification head are allowed to adapt to the pneumonia detection problem. Importantly, this method will directly deal with the training collapse issue witnessed during Phase 2. The model has enough capacity to memories the entire training set with only 86 million parameters that are being updated in just 4,172 training images, a ratio of 20,575:1 between parameters and samples. The model with 14 million trainable parameters (a ratio of roughly 3,353:1) is highly expressive when compared to the training set, but the constraint greatly helps to avoid the fact that the model tends to memories and makes the gradient updates more generalizable, i.e. feature adaptation.

C. *Improvement 2: Regularization using Label Smoothing*

Instead of normal cross entropy, we have used label smoothing cross entropy to avoid model overfitting and regularizing the model to reduce test errors. For this purpose, PyTorch built in function nn.CrossEntropyLoss(label_smoothing = 0.1) is used. The true label y's target distribution with K=2 classes and ϵ=0.1 is updated using:

$\hat{p}(k)$ = (1 - ϵ) $\cdot$ 1[k = y] + ϵ/K

where $\hat{p}$(true) = 0.95, $\hat{p}$(false) = 0.05

In this way this soft target distribution makes sure that the cross entropy loss doesn't make the predicted probability of true class to reach a value of 1.0. It never allows the model to fit training labels with indiscriminately large values of the logit because the cross entropy loss cannot take the value 0 (as 0.95 != 1). This is especially significant on such a small dataset as Kermany where loss collapse occurred with the full fine-tuning model (Phase 2). Class weights to deal with imbalance (NORMAL: 1.4929, PNEUMONIA: 0.5071) are kept and label smoothing is maintained. PyTorch's nn.CrossEntropyLoss allows the use of class weights and label smoothing at the same time, and both are enabled when Kermany is being trained. In the case of BloodMNIST which is roughly class-balanced, there is no use of class weighting, but label smoothing is enabled.

D. *Improvement 3: Transform Bug Fix*

Phase 3 found a bug related to data leakage, which was fixed. The training ImageFolder was instantiated with the training transform pipeline (random horizontal flipping, random rotation up to 10, colour jitter) in Phase 2 implementation. This single dataset object was then split into train and validation subsets using random_split of PyTorch. Since a Subset object uses the same transform as its parent dataset, validation images were randomly augmented per epoch, so that validation measures are not reproducible across epochs so after running they break the signal that causes early stopping. The solution is to instantiate twice the training directory as distinct ImageFolder objects one with the full training augmentation transform, and one with the evaluation-only transform (resize + normalise) and use the same index split on both. The augmented dataset's images are used for training and clean dataset's images are used for validation. In this way it is made sure that the validation metrics produced, must point out the predictable inference on the clean images, and generate a more stable and reproducible validation curve over the training data.

E. *Cross-Domain Extension: BloodMNIST*

Layer freezing, label smoothing and corrected validation pipeline is run, in the same way as in phase 3, on the BloodMNIST dataset to find out that how different the methodology of specific proposed training is from the general proposed training for pneumonia detection. The benchmark finds a radically different task: 8-class microscopy-based classification of peripheral blood cells, on color microscopy images (RGB, 28×28 native resolution, upsampled to 224×224 ViT input) instead of grayscale chest radiographs. The classification task is more difficult as well - 8-class multi-label classification with morphologically similar confusable classes- than the binary pneumonia detection problem. Good results on BloodMNIST would be good indication that the training regimen is a broadly applicable methodology to fine-tuning ViTs on small medical datasets.

IV. Experimental Setup

A. *Computing Environment*

Experiments were run on Google Colab with an NVIDIA Tesla T4 GPU. The software environment included PyTorch 2.x (with CUDA support), HuggingFace Transformers (for loading the ViT models and using ViTImageProcessor) and MedMNIST Python package (for BloodMNIST data loading). We set a global random seed (42) for Python (random module), NumPy and PyTorch (including CUDA operations by setting torch.backends.cudnn.deterministic=True) to make sure that all results are fully reproducible.

B. *Datasets*

*1) **Kermany Chest X-Ray Dataset**:* The Kermany Chest X-Ray Pneumonia dataset (Kermany et al., 2018) is freely available on Kaggle, and consists of 5,863 labeled chest X-rays, with images classified into NORMAL (1,583 images) and PNEUMONIA (4,273 images). The data is divided into training, validation and test sets in directories. The validation dataset contains only 16 images and is an unreliable way to select models. This practice changes this to a stratified random split of the training directory into 80% training (4,172 images) and 20% validation (1,044 images). The 624-image test set is set aside and used only for final testing.

There is large class imbalance in the training set: 74% of images are PNEUMONIA (3,104 images) and 26% are NORMAL (1,068 images). This requires explicit measures - class weighting of loss and weighted sampling of batches - to avoid learned bias of the model towards the majority class.

*2) **BloodMNIST**:* BloodMNIST is a dataset in the MedMNIST v2 benchmark (Yang et al., 2023). It is a collection of 17,092 microscopy images of peripheral blood cells, with eight different morphological cell types: basophil, eosinophil, erythroblast, immature granulocytes (IG), lymphocyte, monocyte, neutrophil and platelet. The data comes with pre-split training, validation, and test sets: 11,959, 1,712, and 3,421 images, respectively. The original image size is 28x28 pixels (RGB). The images are resized to 224×224 pixels for ViT models. The class distribution is relatively uniform, with around 240-670 samples per class in the test set.

C. Pre-processing and Augmentation

Images are resized to 224×224 to match the input size of ViT-Base-Patch16-224. The pixel values are normalised using the channel-wise mean (μ = [0.485, 0.456, 0.406]) and standard deviation (σ = [0.229, 0.224, 0.225]) as returned by the HuggingFace ViTImageProcessor. For the grayscale X-ray images (Kermany) the image is replicated three times to match the three channels for normalisation.

Training augmentations (applied only to training data):

- Kermany: Random horizontal flip (p=0.5), random rotation (±10°), colour jitter (brightness ±0.2, contrast ±0.2).
- BloodMNIST: Random horizontal flip (p=0.5), random rotation (±15°), colour jitter (brightness, contrast, saturation ±0.2).

Testing transforms (for validation and test data):

- Resize to 224×224 and normalise only. We do not apply any arbitrary augmentations.

For the Kermany dataset, inverse-frequency class weights are computed as: wNORMAL = N / (2 × NNORMAL) = 5216 / (2 × 1068) ≈ 1.4929, and wPNEUMONIA = N / (2 × NPNEUMONIA) ≈ 0.5071. A WeightedRandomSampler is built with per-sample weights related to their class weight, to ensure balanced mini-batches during training.

D. Model Configuration and Hyper-parameters

Parameter	**Phase 2 (Baseline reproduction)**	**Phase 3 (Proposed method)**
Base model	ViT-Base-Patch16-224	ViT-Base-Patch16-224
Trainable parameters	∼86M (all)	∼14M (last 2 blocks + head)
Frozen layers	None	Embedding + blocks 0–9 of 12
Loss function	CrossEntropyLoss + class weights	CrossEntropyLoss + class weights + LS (ϵ=0.1)
Optimiser	Adam(lr=$1e^{-4}$, wd=$1e^{-4}$)	Adam(lr=$1e^{-4}$, wd=$1e^{-4}$)
LR Scheduler	ReduceLROnPlateau (f=0.5, p=3)	ReduceLROnPlateau (f=0.5, p=3)
Max epochs	30	12
Early stopping	Patience=5 (val F1)	Patience=4 (val F1)
Batch size	32	32
Val transform	Augmented (bug)	Eval-only (fixed)
Dropout (hidden/attn)	0.1 / 0.1	0.1 / 0.1

TABLE I
COMPLETE MODEL CONFIGURATION COMPARISON BETWEEN PHASE 2 (BASELINE REPRODUCTION) AND PHASE 3 (PROPOSED METHOD)

E. Evaluation Metrics

For binary classification (Kermany) following metrics were used to evaluate the test set:

- **Accuracy:** Number of samples correctly classified.
- **Recall (Sensitivity):** True Positive Rate (TPR) - proportion of PNEUMONIA samples found. The most important metric from a clinical point of view because false negatives (missed pneumonia) are more risky.
- **F1-Score:** Precision-Recall trade-off weighted harmonic mean, with binary mode used for the PNEUMONIA class.
- **AUC-ROC:** Area Under the Receiver Operating Characteristic Curve, an overall measure of the distinctive power of the model with all classification thresholds.

All Phase 3 Kermany test metrics provide bootstrap confidence intervals (1,000 resamples with replacement, 95% confidence level) to measure the amount of estimation uncertainty due to test set finite size. For the multi-class classification (BloodMNIST), Precision, Recall, F1-Score and one-vs-rest macro AUC-ROC are macro-averaged and reported for each class.

V. RESULTS

A. Phase 2: Kermany Dataset: Reproduced Results

The Phase 2 baseline reproduced the results of Singh et al. (2024) with some improvements to the original paper: we used a proper split (1,044 images) for validation (instead of 16 images), class-weighted cross-entropy loss, WeightedRandomSampler, ReduceLROnPlateau scheduler, and early stopping on validation F1. The model was trained for 6 epochs (until early stopping) and the best checkpoint was saved at epoch 1 (validation F1 = 0.9760). Training loss continually improved from 0.1759 at epoch 1 to 0.0376 at epoch 6.

Metric	Singh et al. 2024	Phase 2	Difference
Accuracy	97.61%	91.67%	-5.94%
Precision	N/A	94.01%	–
Recall	95.00%	92.56%	-2.44%
Specificity	98.00%	90.17%	-7.83%
F1-Score	N/A	93.28%	–
AUC-ROC	0.96	0.97	+0.01

TABLE II
PHASE 2 REPRODUCED RESULTS VS ORIGINAL PAPER

The accuracy of 91.67% is 5.94% lower than in the paper (97.61%). The higher AUC-ROC of 0.97 (vs 0.96 in the paper) confirms good distinctive performance. This accuracy drop can be explained by unknown hyperparameters and splits of the original paper.

Epoch-level training statistics for Phase 2:

Epoch	**Train Loss**	**Val Loss**	**Val Acc (%)**	**Val F1**
$\star$1(Best)	0.1759	0.0956	96.55	0.9760
2	0.0873	0.2675	87.44	0.9057
3	0.0462	0.0826	96.36	0.9745
4	0.0384	0.1452	93.48	0.9532
5	0.0357	0.1577	94.15	0.9582
6 (Early Stop)	0.0376	0.0982	96.07	0.9724

TABLE III
PHASE 2 EPOCH-LEVEL TRAINING STATISTICS

B. Proposed Method - Phase 3 (Kermany)

The Kermany dataset was trained on the proposed Phase 3 model under 9 epochs when it early stopped (patience=4). The highest validation F1 (0.9527) was reached at epoch 5.

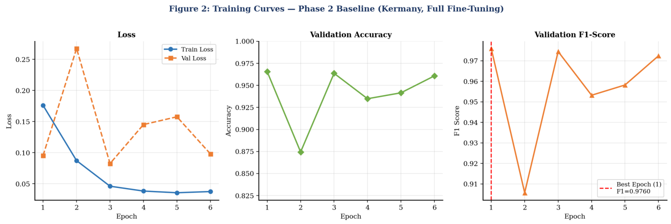

Fig. 2. Training Curves Phase 2

The main difference to Phase 2 can be right away observed in the training loss trend: unlike in Phase 2 training loss which fell down to 0.1759 to 0.0376 (close to zero), Phase 3 training loss fell down progressively to 0.2872 to 0.2148 (never in the locality of zero) which directly proves that the training collapse failure mode was avoided by layer freezing and label smoothing.

Epoch	Train Loss	Val Loss	Val Acc (%)	Val F1
1	0.2872	0.4357	86.30	0.8984
2	0.2532	0.4457	85.63	0.8926
3	0.2337	0.4240	84.58	0.8838
4	0.2237	0.4662	83.43	0.8740
5 ⋆ (Best)	0.2203	0.3564	93.30	0.9527
6	0.2182	0.4677	84.67	0.8846
7	0.2211	0.3866	91.76	0.9412
8	0.2148	0.3806	91.28	0.9375
9 (Early Stop)	0.2156	0.3848	92.05	0.9433

TABLE IV
PHASE 3 KERMANY TRAINING LOG. ⋆ = BEST EPOCH (MODEL CHECKPOINT SAVED). TRAINING LOSS NEVER APPROACHES ZERO, CONFIRMING SUCCESSFUL PREVENTION OF TRAINING COLLAPSE.

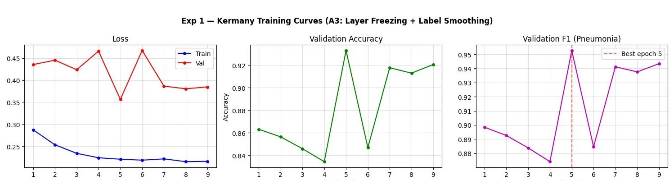

Fig. 3. Training Curves Phase 3

C. Phase 3 vs. Phase 2: Test Set Comparison (Kermany)

Metric	Singh et al. 2024	Phase 2	Phase 3	Change vs P2
Accuracy	97.61%	91.67%	89.90%	-1.77%
Recall (Sensitivity)	95.00%	92.56%	94.87%	+2.31% ↑
Specificity	98.00%	90.17%	81.62%	-8.55%
F1-Score	N/A	93.28%	92.15%	-1.13%
AUC-ROC	0.96	0.97	0.9487	-0.021

TABLE V
FULL COMPARISON ACROSS SINGH ET AL. (2024), PHASE 2 (REPRODUCTION), AND PHASE 3 (PROPOSED METHOD) ON THE KERMANY TEST SET (624 IMAGES). ↑ INDICATES IMPROVEMENT RELATIVE TO PHASE 2.

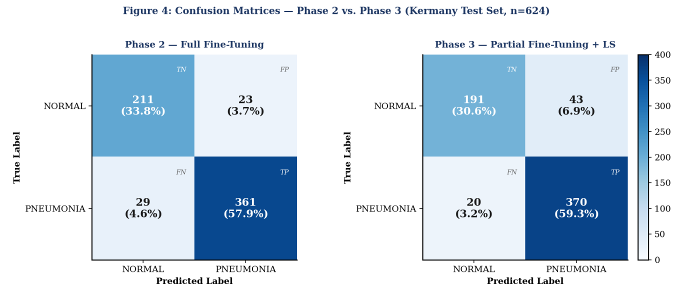

Fig. 4. Confusion Matrices (P2 vs P3 side by side)

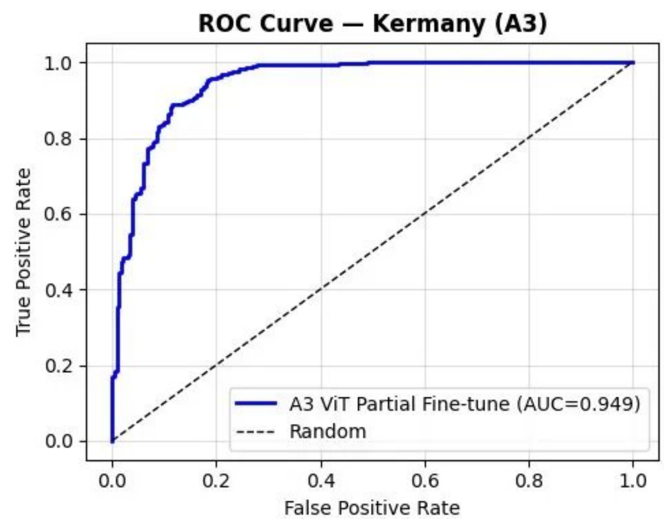

Fig. 5. ROC Curves (P2 vs P3 overlaid

D. Bootstrap Confidence Intervals (Phase 3, Kermany)

Bootstrap resamples were done (1,000 bootstrap resamples) to quantify the uncertainty in Phase 3 test metrics that come due to the finite size of the 624-image test set. The mean, and 95% confidence intervals of each metric are reported in Table 6.

Metric	Mean	95% CI Lower	95% CI Upper	CI Width
Accuracy (%)	89.83	87.50	92.31	4.81
Recall (%)	94.80	92.43	96.95	4.52
Specificity (%)	81.55	76.49	86.49	10.00
F1-Score (%)	92.09	90.12	94.09	3.97
AUC-ROC	0.9483	0.9300	0.9667	0.037

TABLE VI
BOOTSTRAP 95% CONFIDENCE INTERVALS FOR PHASE 3 KERMANY TEST METRICS (1,000 RESAMPLES OF $n = 624$ TEST SET). WIDE CI FOR SPECIFICITY REFLECTS THE SMALLER NORMAL CLASS SUPPORT ($n = 234$).

E. Transfer Learning: BloodMNIST (8-Class Classification)

Phase 3 was executed using the same settings for BloodMNIST. The training was done until the 12 allowed epochs had expired, with continuous improvement thereafter with the highest validation macro F1 reaching 0.9096 on epoch 12.

Epoch	Train Loss	Val Loss	Val Accuracy	Val F1 (Macro)
1	0.8502	0.9039	80.32%	0.7532
2	0.6881	0.8191	84.64%	0.8253
3	0.6593	0.7393	88.32%	0.8741
5	0.6261	0.7257	88.79%	0.8766
8	0.5998	0.6899	89.60%	0.8791
10	0.5925	0.6629	91.12%	0.9002
12 $\star$ (Best)	0.5831	0.6937	91.30%	0.9096

TABLE VII
BLOODMNIST TRAINING LOG (SELECTED EPOCHS). $\star$ = BEST EPOCH. NO EARLY STOPPING TRIGGERED WITHIN 12 EPOCHS.

Metric	Result	Averaging Mode
Accuracy	90.30%	Overall
Precision	90.36%	Macro
Recall	88.82%	Macro
F1-Score	89.25%	Macro
AUC-ROC	0.9917	One-vs-Rest Macro

TABLE VIII
BLOODMNIST TEST SET SUMMARY RESULTS (8-CLASS CLASSIFICATION, $n = 3{,}421$ TEST IMAGES).

Cell Type	Precision	Recall	F1-Score	Test Support
Basophil	0.89	0.85	0.87	244
Eosinophil	0.95	0.98	0.97	624
Erythroblast	0.92	0.91	0.91	311
Immature Granulocytes	0.85	0.76	0.80	579
Lymphocyte	0.85	0.93	0.89	243
Monocyte	0.92	0.69	0.79	284
Neutrophil	0.85	0.99	0.92	666
Platelet	0.99	0.99	0.99	470
Macro Average	0.90	0.89	0.89	3421

TABLE IX
PER-CLASS PRECISION, RECALL, AND F1-SCORE FOR BLOODMNIST TEST SET (8-CLASS CLASSIFICATION).

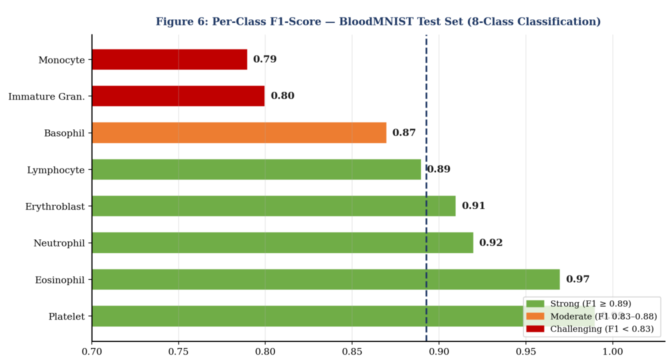

Fig. 6. Per-Class F1 Bar Chart (BloodMNIST)

VI. DISCUSSION

A. *Major Result: Layer Freezing Preventing Training collapse*

The key finding of Phase 3 is that the central hypothesis is actually correct: partial freezing of layers in the context of partial fine-tuning can actually avoid training collapse on the small Kermany dataset. Phase 2 Stage 1 , initial training with no class-weighted sampling and careful regularization showed a training loss of around zero after 5 epochs as well as training F1 of around 1.0 - another obvious indicator of memorization. Even successful Phase 2 run revealed a sharp training loss reduction (0.1759 $\rightarrow$ 0.0376 in six epochs), which indicates a model that was quickly reaching the limits of its performance on the training set.

The qualitative behavior of the training loss curve in Phase 3 (84% of parameters are frozen) is also different: it has a monotonically decreasing curve that does not approach 0.2872, but instead decreases slowly and gradually to 0.2148 in the 9 epochs. This monotonically increasing gradient descent means that the model is progressively learning without exhausting its memory on training set memorization. This is further boosted by the label smoothing which increases the minimum loss possible to more than zero, thus providing the optimizer with meaningful gradient signal during training.

The practical implications of this finding are that medical imaging specialists handling small labeled data would be able to use it. The widely-known suggestion of fully fine-tuning large pretrained models is actively damaging in the case of training datasets with a few thousand samples. The Phase 3 findings indicate that a traditional partial fine-tuning approach in which all but the last encoder blocks are frozen is the default strategy to take when medical image classification tasks have limited annotated data.

B. *Explaining the Kermany Metric Trade-Off.*

In Kermany, Phase 3 results indicate a mixed pattern compared to Phase 2: the recall was improved significantly (2.31% increase, between 92.56% and 94.87%), and specificity was reduced (8.55% decrease, between 90.17% and 81.62%), and accuracy and F1 also fell 1.77% and 1. The pattern itself is internally consistent, and there is a definite systematic explanation of it.

Freezing 10 out of 12 encoder blocks significantly decreases the ability of a model to model the distribution of the training data. With the extreme imbalance of the classes in the Kermany training set (74% pneumonia), the classification capacity is implicitly assigned by gradient descent to the majority of the classes, i.e., the model decision boundary is moved in the direction that minimizes the loss as much as possible, i.e. towards higher sensitivity at the cost of specificity. The outcome is that there is a model that both correctly identifies more cases of pneumonia (higher recall) and also incorrectly identifies more cases of normal as pneumonia (higher false positive rate, lower specificity). This interpretation is verified by the confusion matrix: false positives (NORMAL) which is PNEUMONIA also grow; and false negatives (PNEUMONIA missed) also decrease.

Clinically, this trade-off works in a screening application. The asymmetric cost of errors is in favor of the recall in a screening context: a false negative (unnoticed pneumonia) may leave a patient with an untreated infection that could become more severe; a false positive will result in further clinical examination that can clear up the uncertainty without necessarily causing harm to the patient. The Phase 3 model had a recall of 94.87% (recalling 2.31% higher true pneumonia cases than Phase 2) which is a clinically significant increase in screening sensitivity.

It is also important to add that this small accuracy and F1 decline could be in part due to the fact that the Phase 2 model is more prone to overconfident predictions which coincidentally match the test set distribution, in addition to the true bigger generalization. The non-collapsing and stable training curve of the Phase 3 model offers a better cue that the model has acquired generalizable characteristics and not the patterns of the training-sets memorized.

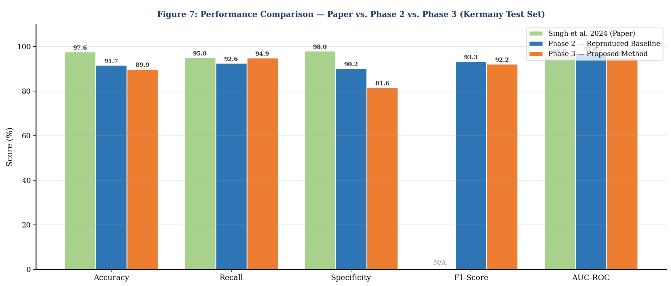

Fig. 7. Overall Metric Comparison Bar Chart

C. Cross-Domain Generalisation: BloodMNIST Analysis.

The results of the BloodMNIST, 90.30 overall accuracy, 89.25 macro F1 and 0.9917 AUC-ROC with 8 cell types, have strong evidence that the proposed Phase 3 training methodology does not depend on the pneumonia detection task but is a generalized way of fine-tuning ViTs on medical imaging datasets.

Major point of attention is the AUC-ROC of 0.9917 (one-vs-rest, macro average). This close to perfect distinction means that when the choice of threshold is not limited, the model can be used to rank cells according to their probability of being in each class in all 8 categories almost perfectly. This large AUC and a smaller macro F1 (0.89) indicates the difficulty of not being able to distinguish between some morphologically similar cell types at the fixed classification threshold (argmax).

Table 9 per-class breakdown indicates that there is a very obvious performance hierarchy that follows to structural interpretability:

- Classes with a high degree of distinctiveness (F1 $\geq$ 0.97): Platelet (F1 =0.99) and Eosinophil (F1=0.97) are visually very distinctive, platelets being small fragments with no nucleus, whereas eosinophils have a typical bilobed nucleus and prominent pink granules. These are discrimination tasks that are easy.
- Moderately unique classes (F1 = 0.890.92): well-classified are Neutrophil (0.92), Erythroblast (0.91) and Lymphocyte (0.89). Neutrophils possess multilobed nuclei, erythroblasts are precursors of red blood cells with a nucleus, and lymphocytes are comparatively small and possess large nucleus.

Structurally ambiguous classes (F1 $\leq$ 0.80): Monocyte (0.79) and Immature Granulocytes (0.80) are the most difficult. The kidney-shaped nuclei of monocytes may resemble activated lymphocytes and immature granulocytes are a heterogeneous developmental spectrum of granulocyte precursors with morphological features that overlap.

This is a non-random failure pattern, the model is unable to learn specifically the sort of cells that are problematic in automated differential counts to human haematologists, and makes it possible that the model is not learning arbitrary image pieces but instead learning clinically relevant discriminative features.

D. Why the Fix of the Validation Transform is Important.

Although it is not explicitly shown in the tables of the test metrics, the validation transform error correction is an important methodological enhancement the effects of which are obvious in the dynamics of the training. During Phase 2, validation F1 curve varied significantly across epochs (between 0.9057 at epoch 2 and 0.9760 at epoch 1 and back to 0.9745 at epoch 3) and it was challenging to tell whether the variation was due to true improvement in the model or was simply the arbitrary effect of the various random augmentations applied to the validation images. Validation metrics are deterministic during each epoch with the bug repaired in Phase 3. The Phase 3 validation curves are thus a solid foundation to make early stopping decisions: when the model actually no longer learns anything on the validation set, early stopping is making the right decision. The scientific reliability of such a study requires this reproducibility, a result of a study that requires a stochastically corrupted validation signal cannot be reproduced, and its statistical analysis cannot be done.

E. Limitation and Future Work.

A number of limitations of this work must be mentioned that refer to the future directions of research.

- **Small training set:** There are around 4,172 training images in the Kermany dataset. The ratio of the number of trainable parameters to the number of samples is also rather high even with 14 million trainable parameters. To further enhance generalisation, adding more labelled data, more dominant augmentation methods (e.g., CutMix, MixUp, RandAugment), or generative data augmentation with diffusion models might be advantageous.
- **Fixed freeze depth:** It was practically determined that the decision to freeze 10 out of 12 encoder blocks was based on the transfer learning theory but was not practically proved in a systematic ablation study. A sweep over freeze depths (freezing 6, 8, 10, and 12 blocks) would determine the best capacity-generalisation trade-off to this particular dataset size and might indicate non-monotonic behaviour with capacity variations.

- **BloodMNIST upscaling:** Original BloodMNIST images of size 28×28 pixels are upscaled 8 times to 224x224 images to match ViT input resolution. This adds in artificial interpolation objects. More architecture friendly would be variants of ViT that are trained on lower-resolution inputs, or variants that apply ViT with smaller patch sizes (e.g. Patch8).
- **No statistical significance test between phases:** Bootstrap CIs are ambiguity measures in each stage individually but are not able to determine that Phase 3 values are statistically different than Phase 2 values, as they are both tested on the same fixed test set. This comparison could be dignified by a paired bootstrap permutation test or a classification error test by McNermar.
- **Label smoothing ϵ ablation:** The default value of ϵ=0.1 was picked according to the literature. An e consisting of 0.05, 0.1, 0.15, 0.2 would determine the best regularization strength when using this dataset and this model configuration.
- **Attention Visualisation**: Qualitative interpretability: Visualisations of attention maps (e.g., with Attention Rollout or GradCAM modified to work with Transformers) would enhance clinical trust in the model by showing that regions of high attention are structurally relevant lung locations related to the pathology of pneumonia.

VII. Conclusion

The paper presents a deep learning pipeline for automated pneumonia detection from chest X-ray images using a Vision Transformer (ViT-Base-Patch16-224), consisting of 3 phases. The main demonstration of this work is that partial fine tuning through selective layer freezing in an effective strategy for adapting large pre trained vision models to small medical imaging datasets.

The baseline of Singh et al. (2024) revealed a fundamental weakness: fully fine tuning the 86 million ViT parameters on approximately 4,172 training images, leading to overfitting where the model memorizes the training set, instead of learning generalizable features. This manifested a nearly zero training loss and perfect f1 score but failed in terms of generalization. To counter this, three targeted interventions were introduced in phase 3. Firstly, freezing the patch embeddings and the first 10 out of 12 reduced trainable parameters from 84 million to 14 million, reducing it by 84%. This directly prevented memorization driven collapse. Secondly, label smoothing with epsilon as 0.1 ensured soft target distribution, preventing overconfident predictions and providing a consistent gradient.

On the Kermany dataset a recall of 94.87% was achieved, surpassing phase 2 by a significant 2.31%. The accompanying a decrease on overall specificity and accuracy, represents a systematic shift towards sensitivity which is justified in medical screening images where undetected pneumonia poses a greater threat than a false alarm. The stable training curves in Phase 3 represent genuine generalization rather than overfitting behaviour observed in Phase 2.

Through cross domain evaluation on the BloodMNIST dataset, an 8 class blood cells classification dataset, the generalizability of the proposed methodology was further validated through an achieved accuracy of 90.30% and AUC-ROC of 0.991. This indicated that the training methodology transfers across imaging modalities and classification tasks.

In summary, this work demonstrates that the common assumption of full fine tuning being the optimal approach is not always applicable, especially when the dataset size is small. When labelled data is scarce, which is often the case in medical imaging, conservative partial fine tuning along with output regularization yields more stable dynamics and a trustworthy model. Future work should explore ablation of freeze depth and alternate strategies along with attention visualization to further validate through visual interpretability.

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